

Optimal Test – Neyman Pearson Lemma

Recall : Power of a test

$$\text{power} = 1 - \beta$$

x	0	1	2	3	4	5
$f(x \theta_0)$	.02	.03	.05	.05	.35	.50
$f(x \theta_1)$	.04	.05	.08	.12	.41	.30

$$H_0 : \theta = \theta_0. \quad H_1 : \theta = \theta_1$$

Test procedure: Decide on the value of a single sample

$$\alpha = 0.05$$

Find A_0 and A_1 .

A_0 : critical region or rejection region

A_1 : complement of critical region

Ack: Hoel, Port, Stone

Recall : Power of a test

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A_0 : critical region or rejection region

A_1 : complement of critical region

A_0	A_1	β
$\{0, 1\}$	$\{2, 3, 4, 5\}$	0.91
$\{2\}$	$\{0, 1, 3, 4, 5\}$	0.92
$\{3\}$	$\{0, 1, 2, 4, 5\}$	0.88

👉 $\alpha = 0.05$

Best Test : $A_0 = \{3\}$

We can compare between tests keeping the significance level i.e. α fixed.

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma

Theorem 7.2.1 (Neyman–Pearson Lemma) *Suppose that one wants to test a simple hypothesis $H_0 : \theta = \theta_0$ versus the simple alternative hypothesis $H_a : \theta = \theta_1$ based on a random sample $X_1, \dots, X_n$ from a distribution with parameter θ . Let $L(\theta) \equiv L(\theta; X_1, \dots, X_n) > 0$ denote the likelihood of the sample when the value of the parameter is θ . If there exist a positive constant K and a subset C of the sample space $\mathbb{R}^n$ (the Euclidean n -space) such that*

1. $\frac{L(\theta_0)}{L(\theta_1)} \leq K$ for $(x_1, x_2, \dots, x_n) \in C$
2. $\frac{L(\theta_0)}{L(\theta_1)} \geq K$ for $(x_1, x_2, \dots, x_n) \in C'$, where C' is the complement of C , and
3. $P[(X_1, \dots, X_n) \in C; \theta_0] = \alpha$.

Then the test with critical region C will be the most powerful test for H_0 versus H_a . We call α the size of the test and C the best critical region of size α .

About the statement :

1. The above statement is about *existence* of a test.
2. The three conditions that implies the lemma.

$$\frac{L(\theta_0)}{L(\theta_1)} \leq K \text{ for } (x_1, x_2, \dots, x_n) \in C$$

$$\equiv \frac{\prod_{i=1}^n f(x_i|\theta_1)}{\prod_{i=1}^n f(x_i|\theta_0)} \geq K' \text{ for } (x_1, x_2, \dots, x_n) \in C$$

$$\frac{L(\theta_0)}{L(\theta_1)} \geq K \text{ for } (x_1, x_2, \dots, x_n) \in C'$$

$$\equiv \frac{\prod_{i=1}^n f(x_i|\theta_1)}{\prod_{i=1}^n f(x_i|\theta_0)} \leq K' \text{ for } (x_1, x_2, \dots, x_n) \in C'$$

$$P[(X_1, X_2, \dots, X_n; \theta_0)] = \alpha$$

N.B. :

- Sample space : $\mathbb{R}^n$
- Random Sample : $X_1, X_2, \dots, X_n$
- $L(\theta) \equiv L(X_1, X_2, \dots, X_n; \theta) \leftarrow$ likelihood of the sample when parameter is θ

Neyman – Pearson Lemma : Application

$$H_0 : \mu_0 \quad H_a : \mu_1 ; \mu_1 > \mu_0$$

Assumptions: Normal population with known variance σ

Likelihood function:

$$L = \prod_{i=1}^n \frac{\exp \left[-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right]}{\sqrt{2\pi}\sigma} = \frac{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right]}{(2\pi\sigma^2)^{n/2}}.$$

Simplification of $\frac{L_1}{L_0} \geq k$ gives an equivalent and simplified definition of critical region

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma : Application

$\frac{L_1}{L_0}$ Calculation :

$$\begin{aligned} (9) \quad \frac{L_1}{L_0} &= \frac{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu_1)^2 \right]}{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right]} \\ &= \exp \left[-\frac{1}{2\sigma^2} \left[\sum (x_i - \mu_1)^2 - \sum (x_i - \mu_0)^2 \right] \right] \\ &= \exp \left[-\frac{1}{2\sigma^2} \left[2(\mu_0 - \mu_1) \sum x_i + n(\mu_1^2 - \mu_0^2) \right] \right] \\ &= \exp \left[\frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i + \frac{n(\mu_0^2 - \mu_1^2)}{2\sigma^2} \right]. \end{aligned}$$

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Neyman – Pearson Lemma : Application

Applying NP lemma:

$$\exp \left[\frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i + \frac{n(\mu_0^2 - \mu_1^2)}{2\sigma^2} \right] \geq k,$$

Taking logarithm of both sides:

$$\frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i \geq \log k + \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}$$

Since $\mu_1 > \mu_0$:

$$\sum x_i \geq \frac{\sigma^2 \log k}{\mu_1 - \mu_0} + \frac{n(\mu_1 + \mu_0)}{2}$$

👉 We reject H_0 if sample mean is greater than certain value--- Our old strategy!

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Neyman – Pearson Lemma : Few Takeaways

$$\sum x_i \geq \frac{\sigma^2 \log k}{\mu_1 - \mu_0} + \frac{n(\mu_1 + \mu_0)}{2}$$

an equivalent and simplified
definition of critical region

leads to definition of test statistic

But k is yet to be
calculated!

$$\frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i \geq \log k + \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}$$

What happens when this is -ve?

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Neyman – Pearson Lemma : Size of critical region

$$\sum x_i \geq \frac{\sigma^2 \log k}{\mu_1 - \mu_0} + \frac{n(\mu_1 + \mu_0)}{2} \quad \rho$$
$$C = \left\{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n x_i \geq \rho \right\}$$

One of the first two conditions give the shape of the region
but NOT size

Size of the region depends on α :

Let $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$; $Z \sim N(0, 1)$; $P(Z > \frac{\rho' - \mu_0}{\sigma/\sqrt{n}}) = 1 - \Phi(\rho'') = 0.05$ where $\rho'' = \frac{\rho' - \mu_0}{\sigma/\sqrt{n}}$, $\rho' = \rho/n$

From normal table : $\rho'' = 1.645$

Hence $\rho' = \mu_0 + 1.645 \sigma/\sqrt{n}$

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Neyman – Pearson Lemma : How to use

1. Determine the likelihood function of alternate hypotheses and null hypotheses.
2. Simplify the inequality $\frac{L_1}{L_0} \geq k$. This gives the critical region. But in this step value of k is still unknown.
3. Then compute k by assuming size of critical region or Type I error to be some given α .

After these three steps we fulfill all the conditions of NP lemma. As a result we get the most powerful test.

Neyman – Pearson Lemma : Example

Example 7.2.2

Let $X_1, \dots, X_n$ denote an independent random sample from a population with a Poisson distribution with mean λ . Derive the most powerful test for testing $H_0 : \lambda = 2$ versus $H_a : \lambda = 1/2$.

Solution

Recall that the pdf of Poisson variable is

$$p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \lambda > 0, x = 0, 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the likelihood function is

$$L = \frac{\left[\lambda^{\sum_{i=1}^n x_i} e^{-\lambda n} \right]}{\prod_{i=1}^n (x_i!)}.$$

$$\frac{L(\theta_0)}{L(\theta_1)} = \frac{\left(2(\sum x_i) e^{-n2}\right)}{\left(\frac{1}{2}\right)^{\sum x_i} e^{-\frac{n}{2}}} < K$$

which implies

$$(4)^{\sum x_i} \left(e^{-\frac{3n}{2}}\right) < K$$

or, taking natural logarithm,

$$\left(\sum x_i\right) \ln 4 - \frac{3n}{2} < \ln K.$$

Solving for $(\sum x_i)$ and letting $\{[\ln K + (3n/2)]/\ln 4\} = K'$, we will reject H_0 whenever $(\sum x_i) < K'$.

Neyman – Pearson Lemma : Example

$$H_0 : p_0 \quad H_a : p_1 ; p_1 > p_0$$

Assumptions: Bernoulli population with known variance σ

Find the test.

Neyman – Pearson Lemma

Neyman-Pearson Lemma *If there exists a critical region C of size α and a nonnegative constant k such that*

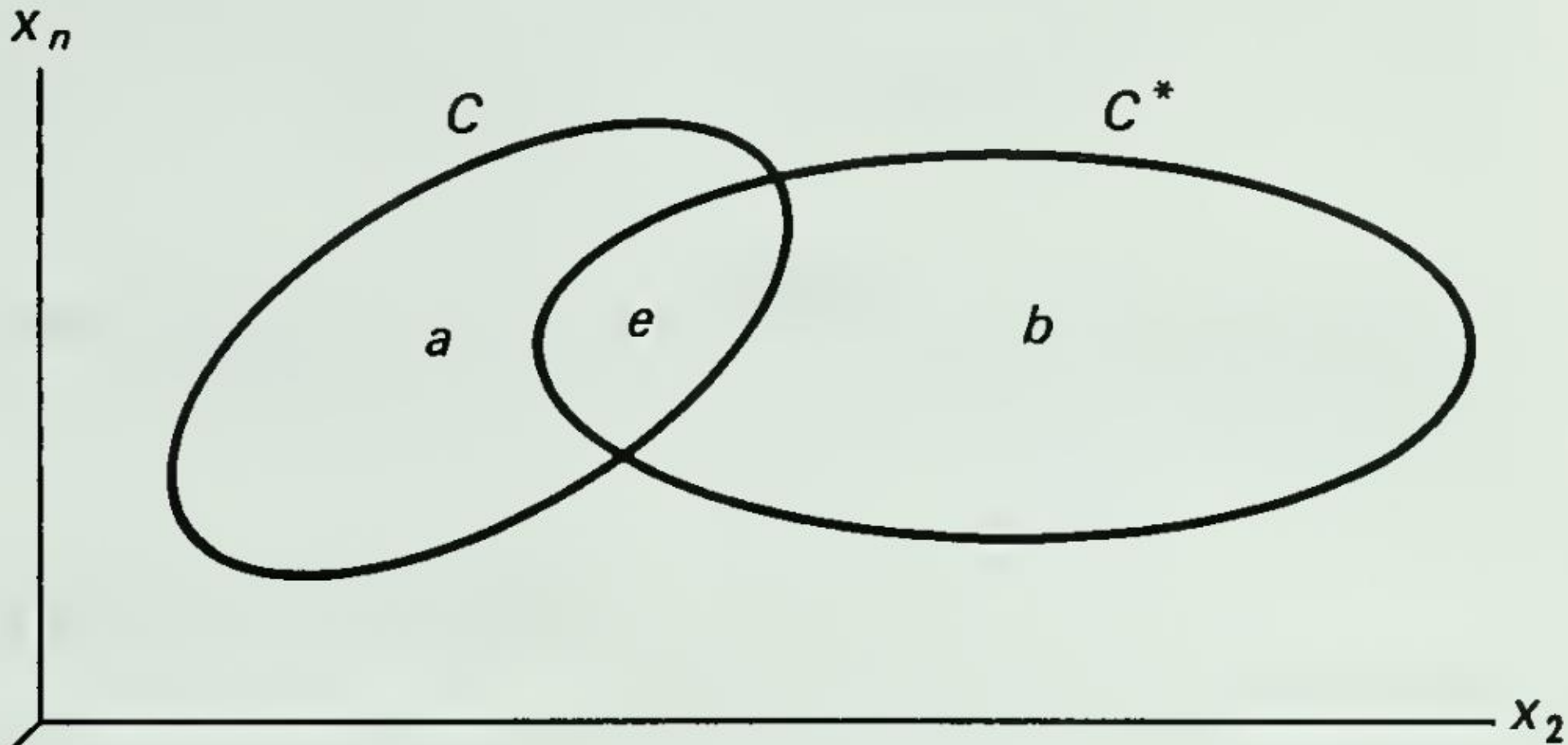
$$(3) \quad \frac{\prod_{i=1}^n f(x_i | \theta_1)}{\prod_{i=1}^n f(x_i | \theta_0)} \geq k \quad \text{for points in } C$$

and

$$\frac{\prod_{i=1}^n f(x_i | \theta_1)}{\prod_{i=1}^n f(x_i | \theta_0)} \leq k \quad \text{for points not in } C,$$

then C is a best critical region of size α .

Hoel, Port,
Stone



Neyman – Pearson Lemma : Proof

Figure 2

An outline of proof :

follows from (1) and (2) that

$$(4) \quad \int_C \cdots \int \prod_{i=1}^n f(x_i | \theta_0) dx_1 \cdots dx_n = \int_C L_0(x) dx = \alpha$$

and

$$\int_{C^*} \cdots \int \prod_{i=1}^n f(x_i | \theta_0) dx_1 \cdots dx_n = \int_{C^*} L_0(x) dx \leq \alpha.$$

Hence

$$\beta = 1 - \int_C L_1(x) dx$$

and

$$\beta^* = 1 - \int_{C^*} L_1(x) dx.$$

Hence,

$$\beta^* - \beta = \int_C L_1(x) dx - \int_{C^*} L_1(x) dx.$$

An outline of proof :

$$\beta^* - \beta = \int_a L_1(x) dx - \int_b L_1(x) dx.$$

$$\int_a L_1(x) dx \geq k \int_a L_0(x) dx.$$

$$\int_b L_1(x) dx \leq k \int_b L_0(x) dx.$$

☞ Hence the proof !

$$\beta^* - \beta \geq k \int_a L_0(x) dx - k \int_b L_0(x) dx.$$